Convex Perturbation and the Suppression of the Indivisibility Deficit

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Abstract

We study how the Chapman–Kolmogorov (CK) deficit of a column-stochastic two-time family $\{\Gamma(t \leftarrow s)\}_{s \le t}$ on a finite state space responds to convex mixing with a divisible reference family, $\Gamma_g = (1-g)\,\Gamma_0 + g\,\Gamma_{\mathrm{ref}}$, $g \in [0,1]$—the simplest fixed-weight convex perturbation of an indivisible family. The organizing result is an exact identity for the deficit operator of the mixed family, $\Delta(t,s,r;\Gamma_g) = (1-g)\,[X - gK]$, where $X = A - CE$ is the substrate deficit and $K = (C-D)(F-E)$ is a substrate–reference interaction term assembled from the kernels on the three subintervals of the triple; the operator-norm bound $D(t,s,r;\Gamma_g) \le (1-g)\,D(t,s,r;\Gamma_0) + g(1-g)\,M(t,s,r)$ follows. Because the identity is exact, the deficit up to temporal scale δt equals $(1-g)\,S(g)$ for a convex function S; the response need not be monotone in g, can vanish at one interior coupling, and can rebound. This forces a two-threshold formulation of detectability: a first-crossing threshold $g_{\mathrm{first}}(\delta t;\varepsilon)$ and a terminal suppression threshold $g_{\mathrm{term}}(\delta t;\varepsilon)$ beyond which the deficit remains below a detection level ε at every stronger coupling. We prove $0 < g_{\mathrm{first}} \le g_{\mathrm{term}} \le g_1 \le g_0 < 1$ with $g_1 = 1 - \big[D_{\mathrm{CK}} + M_* - \sqrt{(D_{\mathrm{CK}} + M_*)^2 - 4M_*\varepsilon}\big]/(2M_*)$ and $g_0 = 1 - \varepsilon/(D_{\mathrm{CK}} + M_*)$, where D_{CK} is the unperturbed deficit up to scale δt and M_* the interaction scale; the bound g_1 is attained by an explicit configuration, so no bound in D_{CK} and M_* alone improves it. The sufficient couplings tend to 1 as $\varepsilon \to 0$; exact vanishing is guaranteed at $g = 1$ and possible at no more than one interior coupling. A two-state family exhibits the interior zero, the rebound, and, for tolerances below the rebound peak, the separation of the two thresholds at the scale level. Secondary results, stated in the body, are an interval-local stability lower bound on g_{first}, quadratic suppression of the integrated deficit, and a rigorous bound for state-dependent coupling. The families are treated as abstract two-time transition families; no claim is made that the mixture is realized by a particular measurement or control protocol.

Keywords. Indivisible stochastic processes; Chapman–Kolmogorov; non-Markovian dynamics; convex perturbation; deficit suppression; detection threshold.

AMS Subject Classifications. 37A50, 47A55, 47B65, 60J05, 60J25.

1 Introduction

A two-time family $\{\Gamma(t \leftarrow s)\}_{s \le t}$ of column-stochastic matrices on a finite configuration space $\Omega = \{1, \dots, N\}$ encodes transition statistics via $\Gamma_{ij}(t \leftarrow s) = \mathbb{P}(X_t = i \mid X_s = j)$ when the family is induced by a process. The family satisfies the Chapman–Kolmogorov (CK) factorization condition if

$$\Gamma(t \leftarrow s) \;=\; \Gamma(t \leftarrow r)\,\Gamma(r \leftarrow s) \qquad \text{for all } 0 \le s < r < t \le T. \tag{1}$$

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A family that fails (1) on a set of triples of positive measure is *indivisible* in the sense of [5, 3, 4]. The companion paper [1] develops the operator-theoretic structure of indivisible families: the singular geometry of the composition-deficit operator, the local indivisibility tensor, the Stinespring dilation of the empirical kernel, and the causal organization on $L^2([0,T]\times\Omega)$. The present paper imports from it only the notion of a two-time transition family [1, Definition 2.3], the deficit notation, the Markov/CK hierarchy [1, Proposition 2.6, Corollary 2.7], the characterization of divisibility by the integrated deficit [1, Theorem 7.4], and, for one remark, the diagonal-retention lift [1, Definition 5.1]; everything else below is proved here. The foundations paper [2] argues independently that dynamical completeness of the instantaneous state is an assumption rather than a consequence, so that divisibility is the case requiring justification.

This paper asks how the CK deficit of an indivisible family responds when the family is mixed, in convex combination, with a divisible one. The mixture $\Gamma_g = (1-g)\Gamma_0 + g\Gamma_{\mathrm{ref}}$ of Definition 2.6 is the simplest fixed-weight convex perturbation of an indivisible family. It always defines a valid column-stochastic two-time family, and every result below is a statement about that family. We make no claim that a measurement, control, or feedback protocol produces the mixture with a time-independent weight: a global random choice between two dynamics, conditioned at a later time on the state, generally produces state- and time-dependent weights (Section 6 treats that case), and the constant-weight model is adopted for its tractability, not derived. We also do not assume that the families are induced by stochastic processes; they are abstract two-time transition families in the sense of [1, Definition 2.3], and process-induced families are the special case. Mixing with a divisible reference does not simply scale the substrate deficit by the residual mass $1-g$: the exact identity of Theorem 3.1 contains a second term, a substrate–reference interaction of order $g(1-g)$, and the interplay of the two terms organizes the paper.

The contributions are four. First, the exact deficit identity (Theorem 3.1) and its consequence, the pencil form $D_{\mathrm{CK}}(\delta t;\Gamma_g) = (1-g)\,S(g)$ of the deficit up to temporal scale δt, with S convex (Proposition 3.2). Because the identity is exact, any nonmonotonicity of the response is an algebraic fact about the family rather than an artifact of estimation, and the identity characterizes the couplings at which the mixed family becomes divisible: at $g=1$ always, and at an interior coupling only if the substrate deficit is the same fixed multiple of the interaction operator at every triple, which can happen for at most one coupling; equivalently, only if the substrate is an affine extrapolation of a divisible family through the reference (Proposition 4.10). Second, the nonmonotonicity forces two detectability thresholds, a first-crossing threshold g_{first} and a terminal suppression threshold g_{term} beyond which the deficit stays below a detection level ε; Theorem 4.3 proves $0 < g_{\mathrm{first}} \le g_{\mathrm{term}}$ together with an explicit sufficient coupling obtained from the quadratic envelope of the identity, of which the linear bound $1-\varepsilon/(D_{\mathrm{CK}}+M_*)$ is a corollary; the quadratic envelope is attained (Remark 4.5), so the sufficient coupling cannot be improved from D_{CK} and M_* alone. Third, an interval-local stability statement (Proposition 4.8): on any compact band of temporal scales on which the unperturbed deficit stays uniformly above ε, no coupling weaker than an explicit $\eta(\varepsilon) > 0$ destroys detectability; and the integrated deficit of [1, Theorem 7.4] is suppressed at least quadratically in $1-g$ (Theorem 5.1). Fourth, a rigorous finite-dimensional bound for state-dependent coupling (Proposition 6.1). The techniques are a bilinear expansion, the triangle inequality, convexity of a norm, and compactness. The content of the paper is the exact identity and the threshold structure it forces.

Closest literature. The behavior of non-Markovianity under convex combinations of dynamical maps has an established literature in the quantum CPTP setting: Wudarski–Chruściński [18], Jagadish–Srikanth–Petruccione [19], Siudzińska [20], Pathak–Shaji [21], and recently Xu–Jagadish [22], who show that non-Markovianity is not additive under mixing of Weyl maps. Neither convex mixing nor nonmonotonicity under mixing is claimed as new here. The robust-

ness measures of non-Markovianity defined through the minimum amount of Markovian dynamics that must be admixed to render an evolution Markovian are closer: Anand–Brun [23], the constant-weight incoherent-mixing version of De Santis–Giovannetti [24], which minimizes the Markovian weight subject to Markovianity at all times, and the convex resource theory of Bhattacharya–Bhattacharya–Majumdar [25]. The terminal suppression threshold of Section 4 is, conceptually, a fixed-reference, tolerance-relaxed variant of the De Santis–Giovannetti measure, and the originality of this paper lies in the setting and the identity rather than in the idea of Markovian admixture. Four points are specific to this paper: (i) the reference is a fixed divisible family, and the thresholds are in the weight against that reference, not a minimization over admixtures; (ii) the object is the classical two-time CK deficit in operator norm, not CP-divisibility of a quantum dynamical map; (iii) the thresholds concern ε-level suppression up to a temporal scale δt, not exact Markovianization at all times; (iv) the exact identity, whose interaction term factors asymmetrically because one summand is divisible, and the first-crossing/terminal distinction it forces. The relation to divisibility-based non-Markovianity functionals [10, 11, 12, 13, 14, 15, 16, 17], to the process-tensor formalism [26, 5, 27, 28], and to product-form speed-limit inequalities is taken up in Section 7; none of the three is used in any proof.

Outline. Section 2 fixes the framework. Section 3 proves the identity, the pencil form, and the suppression bound. Section 4 defines the two thresholds, proves the terminal suppression theorem and the stability result, works a two-state example through both regimes, and gives a scale-level family with an interior zero and separated thresholds (Proposition 4.10). Section 5 treats the integrated deficit. Section 6 treats state-dependent coupling and monotone reparametrization. Section 7 positions the results and states the scope of the detectability notion.

2 Framework

We follow the conventions of [1]: Δ denotes the deficit *operator* and D its operator norm. The families of Definition 2.1 satisfy the boundary and continuity conditions (R1)–(R2) of [1, Theorem 5.4] by construction, and the uniform bound (C1) and operator-norm continuity (C2) under which [1, Theorem 7.4] characterizes divisibility by the integrated deficit, the bound with constant $\sqrt{N}$ by finite-dimensionality. For orientation only: an indivisible family admits no time-ordered-exponential representation by a generator family in the strongly measurable, Bochner-integrable class on the observed state space [1, Proposition 3.6]; this excludes an ordinary time-local evolution-family representation and nothing more—memory-kernel, Volterra, distributional, or augmented-state descriptions are not excluded, and none is used here.

2.1 Two-time families and the deficit

Let $\Omega = \{1, \dots, N\}$, μ counting measure on Ω, and $\mathcal{H} = L^2(\Omega, \mu) \cong \mathbb{C}^N$ with the standard inner product. A column-stochastic matrix $\Gamma \in \mathbb{R}_{\geq 0}^{N \times N}$ acts on $\mathcal{H}$ by $(\Gamma f)(i) = \sum_j \Gamma_{ij} f(j)$; the column-sum normalization $\sum_i \Gamma_{ij} = 1$ reads $\Gamma^* \mathbf{1} = \mathbf{1}$ with $\mathbf{1} = (1, \dots, 1)$. Probability distributions are column vectors and propagate as $p_t = \Gamma(t \leftarrow s)\, p_s$; many texts use the transposed, row-stochastic convention. Let $[0,T]^2_{\leq} = \{(t,s) : 0 \leq s \leq t \leq T\}$.

Definition 2.1 (Two-time column-stochastic family)**.** A *two-time column-stochastic family* on Ω over horizon $T > 0$ is a map $\Gamma\colon [0,T]^2_{\leq} \to \mathbb{R}_{\geq 0}^{N \times N}$ such that for all $(t,s) \in [0,T]^2_{\leq}$, $\Gamma(t \leftarrow s)$ is column-stochastic, $\Gamma(s \leftarrow s) = I$, and $(t,s) \mapsto \Gamma(t \leftarrow s)$ is continuous in operator norm.

Each column of a column-stochastic matrix has Euclidean norm at most 1, so $\|\Gamma\|_{\mathrm{op}} \le \|\Gamma\|_F \le \sqrt{N}$; differences of two such matrices have operator norm at most $2\sqrt{N}$.

Definition 2.2 (Deficit operator, deficit, and deficit up to scale δt)**.** For a two-time family Γ and a triple (t, s, r) with $0 \le s < r < t \le T$, the *deficit operator* and the *deficit* at (t, s, r) are

$$\Delta(t, s, r; \Gamma) \;=\; \Gamma(t \leftarrow s) - \Gamma(t \leftarrow r)\,\Gamma(r \leftarrow s), \qquad D(t, s, r; \Gamma) \;=\; \|\Delta(t, s, r; \Gamma)\|_{\mathrm{op}}\,. \tag{2}$$

For $\delta t \in (0, T]$, the *deficit up to temporal scale* δt is

$$D_{\mathrm{CK}}(\delta t; \Gamma) \;=\; \sup_{\substack{(t,s,r):\\ s<r<t,\ t-s\le\delta t}} D(t, s, r; \Gamma). \tag{3}$$

The quantity $D_{\mathrm{CK}}(\delta t; \Gamma)$ is a cumulative worst case over all triples of duration at most δt, not a deficit at one exact duration; it is nondecreasing in δt, and we speak of it as the deficit *up to* scale δt (the notation D_{CK} is retained). The family is divisible if and only if $D_{\mathrm{CK}}(\delta t; \Gamma) = 0$ for all $\delta t \in (0, T]$.

Lemma 2.3 (Attainment)**.** *For a two-time family Γ and $\delta t \in (0, T]$, the formula (2) defines an operator-norm continuous map on the compact set*

$$K(\delta t) \;=\; \{(t, s, r) : 0 \le s \le r \le t \le T,\ t - s \le \delta t\},$$

which vanishes on the degenerate faces $r = s$, $r = t$, and $t = s$. Hence the supremum in (3) over strict triples equals the maximum of D over $K(\delta t)$, and is finite and attained.

Proof. Continuity on $K(\delta t)$ follows from norm continuity and uniform boundedness of Γ, since products of uniformly bounded norm-continuous families are norm-continuous. On the faces, $\Gamma(u \leftarrow u) = I$ gives $\Delta = 0$: at $r = s$, $\Delta = \Gamma(t \leftarrow s) - \Gamma(t \leftarrow s)\,I$; at $r = t$, $\Delta = \Gamma(t \leftarrow s) - I\,\Gamma(t \leftarrow s)$; and $t = s$ forces $r = s = t$. The function D is continuous and nonnegative on the compact $K(\delta t)$ and vanishes on the faces. If its maximum is positive, every maximizer lies off the faces and is a strict triple, so the supremum over strict triples equals the maximum; if the maximum is zero, $D \equiv 0$ and both quantities vanish. □

Remark 2.4 (Hierarchy with non-Markovianity; scope). By [1, Proposition 2.6], the Markov property implies CK factorization, but two-time CK factorization does not imply Markovianity: the standard witness is Feller's non-Markovian process with the semigroup property [6], whose two-point transition probabilities compose exactly while the three-time joint laws are inconsistent with any Markov chain sharing them. Indivisibility—failure of CK at the two-time level—is therefore strictly stronger than non-Markovianity [1, Corollary 2.7]. The results of this paper concern the suppression of the two-time CK deficit and nothing else: they neither address nor claim to eliminate temporal memory that is invisible at the two-time level.

2.2 The divisible reference and the convex perturbation

Definition 2.5 (Divisible reference family)**.** A *divisible reference family* on Ω over horizon T is a two-time column-stochastic family Γ_{ref} with $\Gamma_{\mathrm{ref}}(t \leftarrow s) = \Gamma_{\mathrm{ref}}(t \leftarrow r)\,\Gamma_{\mathrm{ref}}(r \leftarrow s)$ for all $0 \le s \le r \le t \le T$.

Divisibility is imposed as a hypothesis, not derived: the transition kernels of any time-inhomogeneous Markov process on Ω with norm-continuous kernel supply examples, and a collection of kernels each conditioned on the present state is *not* in general divisible, since such kernels need not compose along intermediate times. No physical interpretation of the reference is attached to the mathematics.

Definition 2.6 (Convex perturbation)**.** Let Γ_0 be a two-time column-stochastic family and Γ_{ref} a divisible reference family on the same Ω and horizon T. For $g \in [0,1]$, the *convex perturbation at mixing weight* g is the two-time family

$$\Gamma_g(t \leftarrow s) \;=\; (1-g)\,\Gamma_0(t \leftarrow s) \;+\; g\,\Gamma_{\mathrm{ref}}(t \leftarrow s), \qquad (t,s) \in [0,T]^2_{\leq}. \tag{4}$$

The mixture is column-stochastic for each g and inherits the boundary and continuity conditions of Definition 2.1. We call g the mixing weight, or the coupling, throughout.

Remark 2.7 (Status of the model)*.* Equation (4) is a formal perturbation path in the convex set of two-time column-stochastic families, joining the substrate to the reference; it is the simplest fixed-weight model, chosen because the response of the deficit to a single scalar weight can be computed exactly. Three things are *not* claimed. It is not claimed that Γ_g is the transition family of a stochastic process with a compatible hierarchy of finite-dimensional distributions; the theorems are statements about two-time families and need no such realization. It is not claimed that a measurement, control, or feedback intervention produces (4) with a time-independent weight; a random choice between two dynamics made once and conditioned later on the state produces weights that depend on the time and the state, which is the family of Section 6 (Remark 6.2). No interpretation of g beyond that of a mixing weight is used in any statement or proof. Alternatives—generator perturbations, sequential interventions, kernel compositions, and time- or state-dependent weights—are natural; the fixed-weight convex path is the leading case, and Section 6 shows how far its estimates extend.

Remark 2.8 (The perturbation is not a divisibility extension)*.* The convex perturbation does not augment the configuration space or pass to a divisible Markov representation on a larger state space, in the sense of the projection-operator and hidden-Markov constructions [7, 8, 9]. Such augmentations produce a divisible kernel on a larger $\widetilde{\Omega} \supsetneq \Omega$ whose marginal on Ω is the original family; the augmentation eliminates the deficit by construction. The convex perturbation operates on Ω itself, and the deficit of Γ_g on Ω is the object of study.

3 The deficit identity, the pencil form, and the suppression bound

Theorem 3.1 (Deficit identity and suppression bound)**.** *Let Γ_0 be a two-time column-stochastic family and Γ_{ref} a divisible reference family on the same (Ω, T), and let Γ_g be the convex perturbation of Definition 2.6. For a triple $0 \leq s < r < t \leq T$ write $A = \Gamma_0(t \leftarrow s)$, $C = \Gamma_0(t \leftarrow r)$, $E = \Gamma_0(r \leftarrow s)$, $D = \Gamma_{\mathrm{ref}}(t \leftarrow r)$, $F = \Gamma_{\mathrm{ref}}(r \leftarrow s)$, and set*

$$X(t,s,r) \;:=\; A - CE \;=\; \Delta(t,s,r;\Gamma_0), \qquad K(t,s,r) \;:=\; (C-D)(F-E). \tag{5}$$

Then for every $g \in [0,1]$,

$$\Delta(t,s,r;\Gamma_g) \;=\; (1-g)\,X \;-\; g(1-g)\,K \;=\; (1-g)\,\big[X - gK\big], \tag{6}$$

and hence

$$D(t,s,r;\Gamma_g) \;=\; (1-g)\,\|X - gK\|_{\mathrm{op}} \;\leq\; (1-g)\,D(t,s,r;\Gamma_0) + g(1-g)\,M(t,s,r), \qquad M(t,s,r) := \|K(t,s,r)\|_{\mathrm{op}} \tag{7}$$

In particular $g \mapsto \Delta(t,s,r;\Gamma_g)$ is a polynomial in g, $D(t,s,r;\Gamma_g)$ is continuous in g, and $D(t,s,r;\Gamma_1) = 0$.

Proof. Write $B = \Gamma_{\mathrm{ref}}(t \leftarrow s)$; divisibility gives $B = DF$. Expanding,

$$\begin{aligned}\Gamma_g(t \leftarrow s) - \Gamma_g(t \leftarrow r)\,\Gamma_g(r \leftarrow s) &= \big[(1-g)A + g\,DF\big] - \big[(1-g)C + gD\big]\big[(1-g)E + gF\big] \\ &= (1-g)A + g\,DF - (1-g)^2 CE - g(1-g)(CF + DE) - g^2 DF.\end{aligned}$$

Since $g\,DF - g^2 DF = g(1-g)\,DF$ and $(1-g)A = (1-g)^2 A + g(1-g)A$, this equals $(1-g)^2(A-CE) + g(1-g)\big[A + DF - CF - DE\big]$, and the bracket is $(A-CE) - (C-D)(F-E)$ because $(C-D)(F-E) = CF - CE - DF + DE$. Hence the deficit is $[(1-g)^2 + g(1-g)](A - CE) - g(1-g)(C-D)(F-E) = (1-g)(A-CE) - g(1-g)(C-D)(F-E)$, which is (6); factoring $(1-g) \geq 0$ out of the norm gives the equality in (7), and the triangle inequality gives the bound. The remaining assertions are read off from (6). □

The interaction term K is a product of two substrate–reference differences, one on the late subinterval (t, r) and one, with sign reversed, on the early subinterval (r, s). Submultiplicativity gives $M(t,s,r) \leq \|\Gamma_0(t \leftarrow r) - \Gamma_{\mathrm{ref}}(t \leftarrow r)\|_{\mathrm{op}}\, \|\Gamma_0(r \leftarrow s) - \Gamma_{\mathrm{ref}}(r \leftarrow s)\|_{\mathrm{op}}$: in the substrate–reference distance $W(u,v) := \|\Gamma_0(u \leftarrow v) - \Gamma_{\mathrm{ref}}(u \leftarrow v)\|_{\mathrm{op}}$ on the two subintervals, the interaction term is of second order, while the substrate deficit enters the bound at first order in itself.

Proposition 3.2 (Exact response function up to scale δt; interior divisibility)**.** *Fix $\delta t \in (0, T]$ and let $\mathcal{T}(\delta t)$ denote the set of strict triples with $t - s \leq \delta t$. Define*

$$S_{\delta t}(g) \;:=\; \sup_{(t,s,r)\in\mathcal{T}(\delta t)} \|X(t,s,r) - g\,K(t,s,r)\|_{\mathrm{op}}\,, \qquad g \in [0,1]. \tag{8}$$

Then:

(i) $D_{\mathrm{CK}}(\delta t; \Gamma_g) = (1-g)\, S_{\delta t}(g)$ *for every* $g \in [0,1]$.

(ii) $S_{\delta t}$ *is convex on* $[0,1]$*, Lipschitz with constant* $M_*(\delta t) := \sup_{\mathcal{T}(\delta t)} M(t,s,r)$*, satisfies* $S_{\delta t}(0) = D_{\mathrm{CK}}(\delta t; \Gamma_0)$*, and* $S_{\delta t}(g) \leq D_{\mathrm{CK}}(\delta t; \Gamma_0) + g\, M_*(\delta t)$.

(iii) For $g \in [0,1)$*,* $D_{\mathrm{CK}}(\delta t; \Gamma_g) = 0$ *if and only if* $X(t,s,r) = g\,K(t,s,r)$ *for every* $(t,s,r) \in \mathcal{T}(\delta t)$*. If* $D_{\mathrm{CK}}(\delta t; \Gamma_0) > 0$*, at most one* $g \in [0,1)$ *has this property, and then* $g = \|X(t,s,r)\|_{\mathrm{op}} \,/\, \|K(t,s,r)\|_{\mathrm{op}}$ *at every triple with* $X(t,s,r) \neq 0$.

Proof. (i) Take the supremum over $\mathcal{T}(\delta t)$ in the equality of (7); the factor $(1-g) \geq 0$ is uniform in the triple. (ii) For fixed (t,s,r), $g \mapsto \|X - gK\|_{\mathrm{op}}$ is the norm of an affine function of g, hence convex, and $\big|\|X - gK\| - \|X - g'K\|\big| \leq |g - g'|\, \|K\| \leq |g - g'|\, M_*(\delta t)$; suprema of convex functions are convex, and the Lipschitz bound passes to the supremum. $S_{\delta t}(0) = \sup \|X\| = D_{\mathrm{CK}}(\delta t; \Gamma_0)$, and $\|X - gK\| \leq \|X\| + g\,\|K\|$ gives the last inequality. (iii) By (i), for $g < 1$ the deficit vanishes iff $S_{\delta t}(g) = 0$ iff $X = gK$ on $\mathcal{T}(\delta t)$. If $X = gK$ and $X = g'K$ on $\mathcal{T}(\delta t)$ with $X(\tau) \neq 0$ at some triple τ, then $K(\tau) \neq 0$ and $(g - g')K(\tau) = 0$, so $g = g'$; taking norms in $X = gK$ gives the value of g. □

Remark 3.3 (Nonmonotonicity of the response). The bound in (7), $(1-g)D_0 + g(1-g)M$ with $D_0 := D(t,s,r;\Gamma_0)$, has derivative $-D_0 + (1-2g)M$ in g, so it is nonincreasing on all of $[0,1]$ if and only if $D_0 \geq M$; when the interaction term dominates ($M > D_0$) the *bound* rises on a neighborhood of $g = 0$ before the factor $(1-g)$ drives it to zero. Whether the *exact* deficit $(1-g)\,\|X - gK\|$ is monotone is a different question, governed by the alignment of K with X and not by $M > D_0$ alone: if $K = cX$ with $c > 1$ (as in the interaction-dominant regime of the example below, where $c \approx 2.78$), the exact deficit is $(1-g)|1-cg|\;\|X\|$, which decreases to zero at the interior coupling $g = 1/c$ and then rebounds, whereas for $0 < c \leq 1$ the same formula is strictly decreasing on $[0,1]$; an initial rise of the exact deficit requires, by (i)–(ii), that the right derivative of $\|X - gK\|$ at 0 exceed $\|X\|$, which needs $M > D_0$ and an interaction term pointing against X. Xu and Jagadish [22] show that non-Markovianity is not additive under mixing in the quantum setting; the identity exhibits the non-additivity here as the interaction term. The content of the suppression bound is therefore the exact functional form and the vanishing at $g = 1$, not monotonicity in g; the nonmonotonicity makes a first-crossing threshold operationally insufficient and forces the terminal threshold of Section 4.

Corollary 3.4 (Suppression up to scale δt)**.** *For each $\delta t \in (0,T]$ and $g \in [0,1]$,*

$$D_{\mathrm{CK}}(\delta t;\Gamma_g) \;\le\; (1-g)\big[D_{\mathrm{CK}}(\delta t;\Gamma_0) + g\,M_*(\delta t)\big] \;\le\; (1-g)\big[D_{\mathrm{CK}}(\delta t;\Gamma_0) + M_*(\delta t)\big], \tag{9}$$

where $M_*(\delta t)$ is finite and attained by the argument of Lemma 2.3 applied to the norm-continuous map $(t,s,r)\mapsto K(t,s,r)$, which also vanishes on the degenerate faces.

Proof. Proposition 3.2(i)–(ii). □

4 Detection thresholds: terminal suppression and interval-local stability

By Remark 3.3 the response of the deficit to the coupling need not be monotone, so a single "critical coupling" defined as a first crossing does not support the reading one wants from a threshold: a first crossing can, in general, be followed by re-entry above the detection level. The detectability theory therefore runs on two thresholds.

4.1 The two thresholds and the terminal suppression theorem

Definition 4.1 (First-crossing and terminal suppression thresholds)**.** Fix $\delta t \in (0,T]$ with $D_{\mathrm{CK}}(\delta t;\Gamma_0) > 0$ and a detection level $\varepsilon \in (0, D_{\mathrm{CK}}(\delta t;\Gamma_0))$. Define

$$g_{\mathrm{first}}(\delta t;\varepsilon) \;=\; \inf\big\{g\in[0,1] : D_{\mathrm{CK}}(\delta t;\Gamma_g) < \varepsilon\big\}, \tag{10}$$

$$g_{\mathrm{term}}(\delta t;\varepsilon) \;=\; \inf\big\{g\in[0,1] : D_{\mathrm{CK}}(\delta t;\Gamma_h) < \varepsilon \;\text{ for all } h\in[g,1]\big\}. \tag{11}$$

The first-crossing threshold records where the deficit first dips below the detection level; the terminal threshold records where it goes below the level and stays there. Both sets are nonempty (Theorem 4.3(ii)), so both thresholds are well defined, and $g_{\mathrm{first}} \le g_{\mathrm{term}}$ since every g in the terminal set lies in the first-crossing set. Because the sets are defined by strict inequalities, the infimum need not belong to them: the deficit may equal ε at a threshold, and the statements below distinguish behavior at a threshold from behavior strictly above it.

Lemma 4.2 (Lipschitz dependence on the coupling)**.** *For fixed δt, $g\mapsto D_{\mathrm{CK}}(\delta t;\Gamma_g)$ is Lipschitz on $[0,1]$ with constant $D_{\mathrm{CK}}(\delta t;\Gamma_0) + M_*(\delta t)$.*

Proof. From (6), for $g,g'\in[0,1]$ and any triple, $\Delta(t,s,r;\Gamma_g) - \Delta(t,s,r;\Gamma_{g'}) = (g'-g)X - (g-g')\big(1-(g+g')\big)K$, using $g(1-g) - g'(1-g') = (g-g')(1-(g+g'))$. Since $|1-(g+g')|\le 1$, $\big|D(t,s,r;\Gamma_g) - D(t,s,r;\Gamma_{g'})\big| \le |g-g'|\,[D(t,s,r;\Gamma_0) + M(t,s,r)] \le |g-g'|\,[D_{\mathrm{CK}}(\delta t;\Gamma_0) + M_*(\delta t)]$ uniformly over $\mathcal{T}(\delta t)$, and the bound passes to the supremum. □

Theorem 4.3 (Terminal suppression threshold: existence and sufficient couplings)**.** *Under the hypotheses of Definition 4.1, abbreviate $D := D_{\mathrm{CK}}(\delta t;\Gamma_0)$ and $M := M_*(\delta t)$, and define*

$$g_1 \;:=\; \begin{cases} 1 - \dfrac{D + M - \sqrt{(D+M)^2 - 4M\varepsilon}}{2M}, & M > 0, \\[2ex] 1 - \dfrac{\varepsilon}{D}, & M = 0, \end{cases} \qquad g_0 \;:=\; 1 - \frac{\varepsilon}{D+M}. \tag{12}$$

Then:

(i) (Envelopes) For every $g \in [0,1]$, $D_{\mathrm{CK}}(\delta t;\Gamma_g) \le E(g) := (1-g)(D+gM) \le B(g) := (1-g)(D+M)$. *The quadratic envelope* E *satisfies* $E > \varepsilon$ *on* $[0,g_1)$, $E(g_1) = \varepsilon$, *and* E *is strictly decreasing on* $[g_1,1]$*; the linear envelope* B *is strictly decreasing with* $B(g_0) = \varepsilon$*; and* $0 < g_1 \le g_0 < 1$.

(ii) (Terminal suppression) $D_{\mathrm{CK}}(\delta t;\Gamma_g) \le \varepsilon$ *for every* $g \in [g_1,1]$, *and* $D_{\mathrm{CK}}(\delta t;\Gamma_g) < \varepsilon$ *strictly for every* $g \in (g_1,1]$*; the same holds with* g_0 *in place of* g_1.

(iii) (Threshold chain)

$$0 \;<\; g_{\mathrm{first}}(\delta t;\varepsilon) \;\le\; g_{\mathrm{term}}(\delta t;\varepsilon) \;\le\; g_1 \;\le\; g_0 \;=\; 1 - \frac{\varepsilon}{D_{\mathrm{CK}}(\delta t;\Gamma_0) + M_*(\delta t)} \;<\; 1.$$

Proof. (i) The envelopes are Corollary 3.4. For $M > 0$, $E(g) - \varepsilon = -Mg^2 + (M-D)g + (D-\varepsilon)$ is a downward parabola with value $D - \varepsilon > 0$ at $g = 0$ and $-\varepsilon < 0$ at $g = 1$, so it has exactly one root in $(0,1)$. Its roots are $\big[(M-D) \pm \rho\big]/(2M)$ with $\rho := \sqrt{(D+M)^2 - 4M\varepsilon}$, the radicand being positive because $(D+M)^2 - 4M\varepsilon > (D+M)^2 - 4MD = (D-M)^2 \ge 0$ for $\varepsilon < D$; since $\rho^2 - (M-D)^2 = 4M(D-\varepsilon) > 0$, the smaller root is negative and the larger root is $g_1 = [(M-D)+\rho]/(2M) = 1 - (D+M-\rho)/(2M)$. Hence $E > \varepsilon$ on $[0,g_1)$ and $E(g_1) = \varepsilon$. The vertex of the parabola lies at $(M-D)/(2M) \le g_1$, so E is strictly decreasing on $[g_1,1]$. For $M = 0$, $E = B$ is linear and the statements are immediate. B has slope $-(D+M) < 0$ and $B(g_0) = \varepsilon$ by definition of g_0. Finally $E(g_0) \le B(g_0) = \varepsilon$ while $E > \varepsilon$ on $[0,g_1)$, so $g_0 \ge g_1$; $g_1 > 0$ because $E(0) = D > \varepsilon$; and $g_0 < 1$ because $\varepsilon > 0$.

(ii) For $g \ge g_1$, $D_{\mathrm{CK}}(\delta t;\Gamma_g) \le E(g) \le E(g_1) = \varepsilon$ by monotonicity of E on $[g_1,1]$, with strict inequality for $g > g_1$. The statement for g_0 follows from $g_0 \ge g_1$, or directly from B.

(iii) By (ii), every $g \in (g_1,1]$ belongs to the terminal set (11): for all $h \in [g,1] \subset (g_1,1]$, $D_{\mathrm{CK}}(\delta t;\Gamma_h) < \varepsilon$. Hence the terminal set contains $(g_1,1]$ and $g_{\mathrm{term}} \le g_1$; in particular both sets of Definition 4.1 are nonempty and $g_{\mathrm{first}} \le g_{\mathrm{term}}$. Finally, Lemma 4.2 gives $D_{\mathrm{CK}}(\delta t;\Gamma_g) \ge D - g(D+M) > \varepsilon$ for every $g < \eta := (D-\varepsilon)/(D+M)$, so the first-crossing set is contained in $[\eta,1]$ and $g_{\mathrm{first}} \ge \eta > 0$. □

Remark 4.4 (Operational reading; the diagnostic role of the first crossing). Below g_{first} the deficit up to scale δt is at least ε and CK indivisibility is detectable at that tolerance; above g_{term} it is below ε at every stronger coupling, and the mixed family is indistinguishable from a divisible one *according to the CK-deficit statistic at the chosen temporal scale and tolerance* (Section 7.5 states what that does and does not mean). Between the thresholds the response may be nonmonotone, and a first crossing can in general be followed by re-entry above ε; every statement of the form "above this coupling indivisibility is undetectable" therefore refers to g_{term}, never to g_{first}. The first-crossing threshold keeps a diagnostic role, since it is the quantity read off a plotted response, and the example of Section 4.3 reports it; it is nowhere interpreted terminally. The sufficient couplings $g_1 \le g_0$ tend to 1 as $\varepsilon \to 0$: the theorem supplies arbitrarily strong suppression at a sub-unit coupling for each fixed tolerance, not exact vanishing, which by Proposition 3.2(iii) is guaranteed at $g = 1$ and can occur at most at one interior coupling.

Remark 4.5 (Sign-blindness of the bounds). The bounds of Theorem 4.3 use only the two summary statistics $D = D_{\mathrm{CK}}(\delta t;\Gamma_0)$ and $M = M_*(\delta t)$, and they are sign-blind in the interaction term: as M grows, both g_1 and g_0 move toward 1, so a larger interaction term *weakens* the sufficient bound and never shows suppression at smaller coupling. The exact response $(1-g)S_{\delta t}(g)$ of Proposition 3.2 sees the alignment of K with X that the summaries discard: when the interaction term aligns with the substrate deficit, the actual thresholds can be far below the bounds (in the interaction-dominant regime of Section 4.3, $g_{\mathrm{term}} \approx 0.108$ against $g_1 \approx 0.807$ and $g_0 \approx 0.834$). The invariant content is $S_{\delta t}$ itself, the norm of the operator pencil $X - gK$ maximized over triples; the bounds summarize the pencil by two numbers. The

quadratic envelope is attained, so the summary cannot be improved: if a triple $\tau \in \mathcal{T}(\delta t)$ attains both suprema, $\|X(\tau)\| = D$ and $\|K(\tau)\| = M$, and $K(\tau) = -cX(\tau)$ with $c = M/D$, then $\|X(\tau) - gK(\tau)\| = D + gM$ for every g, so $S_{\delta t}(g) = D + gM$ by Proposition 3.2(ii), $D_{\mathrm{CK}}(\delta t; \Gamma_g) = E(g)$ on all of $[0,1]$, and $g_{\mathrm{first}} = g_{\mathrm{term}} = g_1$. Hence no bound expressed in D and M alone improves Theorem 4.3. Section 4.3 realizes this configuration inside the two-state example. When M is small, g_1 and g_0 both reduce to $1 - \varepsilon/D$, the pure linear suppression rate of the convex combination.

4.2 Interval-local stability of detectability

The theorem bounds the thresholds from above. The complementary question is whether an arbitrarily weak coupling can destroy detectability; on a compact band of temporal scales on which the unperturbed deficit stays uniformly above the detection level, it cannot, with an explicit constant. This is a stability statement obtained from uniform separation and continuity on a compact set, and nothing more.

Lemma 4.6 (Joint continuity)**.** *The map $(\delta t, g) \mapsto D_{\mathrm{CK}}(\delta t; \Gamma_g)$ is jointly continuous on $(0, T] \times [0, 1]$, and $\delta t \mapsto M_*(\delta t)$ is continuous on $(0, T]$.*

Proof. The map $(t, s, r, g) \mapsto D(t, s, r; \Gamma_g) = (1-g)\,\|X(t, s, r) - gK(t, s, r)\|$ is jointly continuous on $K(T) \times [0, 1]$, since X and K are norm-continuous (Lemma 2.3) and uniformly bounded. The correspondence $\delta t \mapsto K(\delta t)$ is compact-valued and continuous in the Hausdorff sense on $(0, T]$: it is nested increasing, any triple with $t - s \le \delta t$ is approximated for $\delta t' < \delta t$ by its affine contraction toward the midpoint of $[s, t]$, which moves it by $O(\delta t - \delta t')$, and upper hemicontinuity follows from closedness of the constraint $t - s \le \delta t$. Berge's maximum theorem [37, Ch. VI] (see also [38, Theorem 17.31]) gives continuity of the value function $(\delta t, g) \mapsto \max_{K(\delta t)} D(\cdot; \Gamma_g)$, which equals $D_{\mathrm{CK}}(\delta t; \Gamma_g)$ by Lemma 2.3. The same argument applied to $K(t, s, r)$ gives continuity of M_*. □

Assumption 4.7 (Uniform detectability on a band of scales)**.** Fix a nondegenerate interval $[\delta t_-, \delta t_+] \subset (0, T]$ and $\varepsilon > 0$ with

$$m_0 \;:=\; \min_{\delta t \in [\delta t_-, \delta t_+]} D_{\mathrm{CK}}(\delta t; \Gamma_0) \;>\; \varepsilon,$$

the minimum existing by Lemma 4.6. A sufficient condition is a lower scaling bound $D_{\mathrm{CK}}(\delta t; \Gamma_0) \ge c_\alpha(\delta t)^\alpha$ on the band with $\varepsilon < c_\alpha(\delta t_-)^\alpha$; under the hypotheses of the local scaling law [1, Proposition 4.12, Definition 4.11] the exponent α is governed by the order of vanishing of the indivisibility tensor. The exponent serves only to guarantee the gap $m_0 > \varepsilon$ and plays no further role.

Proposition 4.8 (Interval-local stability of the detection threshold)**.** *Under Assumption 4.7, let $\overline{M} := \max_{[\delta t_-, \delta t_+]} M_*(\delta t)$ and define*

$$\eta(\varepsilon) \;:=\; \begin{cases} \dfrac{m_0 + \overline{M} - \sqrt{(m_0 - \overline{M})^2 + 4\,\overline{M}\varepsilon}}{2\,\overline{M}}, & \overline{M} > 0, \\[2ex] 1 - \dfrac{\varepsilon}{m_0}, & \overline{M} = 0. \end{cases} \tag{13}$$

Then $\eta(\varepsilon) \in (0, 1)$ and $g_{\mathrm{first}}(\delta t; \varepsilon) \ge \eta(\varepsilon)$ for every $\delta t \in [\delta t_-, \delta t_+]$. In particular the product $g_{\mathrm{first}}(\delta t; \varepsilon)\,\delta t$ is bounded below by $\eta(\varepsilon)\,\delta t_-$ on the band, and above by $\delta t\, g_1(\delta t; \varepsilon) \le \delta t_+$; the product form is a restatement of the uniform bound on g_{first} and carries no independent content.

Proof. Fix δt in the band and let (t^*, s^*, r^*) attain $D_{\mathrm{CK}}(\delta t; \Gamma_0)$ (Lemma 2.3; the maximizer is a strict triple because the maximum is positive). By the reverse triangle inequality in (7), for every g,

$$D_{\mathrm{CK}}(\delta t; \Gamma_g) \;\geq\; D(t^*, s^*, r^*; \Gamma_g) \;\geq\; (1-g)\, D_{\mathrm{CK}}(\delta t; \Gamma_0) - g(1-g)\, M(t^*, s^*, r^*) \;\geq\; (1-g)\big[m_0 - g\,\overline{M}\big],$$

using $1 - g \geq 0$ with $D_{\mathrm{CK}}(\delta t; \Gamma_0) \geq m_0$, and $g(1-g) \geq 0$ with $M(t^*, s^*, r^*) \leq M_*(\delta t) \leq \overline{M}$. The right-hand side exceeds ε exactly when $q(g) := \overline{M} g^2 - (m_0 + \overline{M})g + (m_0 - \varepsilon) > 0$. For $\overline{M} > 0$, q is an upward parabola with $q(0) = m_0 - \varepsilon > 0$ and $q(1) = -\varepsilon < 0$; its smaller root is $\eta(\varepsilon)$ (the discriminant is $(m_0 + \overline{M})^2 - 4\overline{M}(m_0 - \varepsilon) = (m_0 - \overline{M})^2 + 4\overline{M}\varepsilon$), it lies in $(0, 1)$, and on $[0, 1]$ one has $q > 0$ precisely on $[0, \eta(\varepsilon))$. For $\overline{M} = 0$ the condition is $(1 - g)m_0 > \varepsilon$, i.e. $g < 1 - \varepsilon/m_0$. In either case $D_{\mathrm{CK}}(\delta t; \Gamma_g) > \varepsilon$ for all $g \in [0, \eta(\varepsilon))$ and all δt in the band, so the first-crossing set is contained in $[\eta(\varepsilon), 1]$. The product statements follow by multiplying by $\delta t \in [\delta t_-, \delta t_+]$ and using Theorem 4.3(iii). □

Remark 4.9 (Scope of the stability result). On a band of scales where the unperturbed deficit is uniformly detectable, no coupling weaker than $\eta(\varepsilon)$ pushes the deficit below the detection level anywhere on the band. The constant is manufactured from the band and the substrate–reference pair; g is a mixing weight, not a rate; no $\delta t \to 0$ asymptotics is analyzed. Section 7.4 compares the product form with speed-limit inequalities and finds the resemblance formal only.

4.3 A two-state illustration

We work an example on $\Omega = \{1, 2\}$ at the single triple $(s, r, t) = (0, 1, 2)$. It verifies the identity, exhibits the substrate-dominant and interaction-dominant regimes including an interior zero and a rebound, shows the thresholds and both bounds, and realizes the attainment configuration of Remark 4.5. Because the example is specified at one triple, the scale-level quantities $D_{\mathrm{CK}}(\delta t; \cdot)$ and $M_*(\delta t)$ are replaced by their pointwise counterparts $D(t, s, r; \cdot)$ and $M(t, s, r)$, and the pencil $S(g)$ by $\|X - gK\|$; the example does not construct a continuous two-time family, and the scale-level suprema are not realized. A scale-level family, with an interior zero and separated thresholds, follows at the end of the subsection (Proposition 4.10).

Substrate and reference. Let

$$E := \Gamma_0(r \leftarrow s) = \begin{pmatrix} 0.8 & 0.2 \\ 0.2 & 0.8 \end{pmatrix}, \quad C := \Gamma_0(t \leftarrow r) = \begin{pmatrix} 0.7 & 0.3 \\ 0.3 & 0.7 \end{pmatrix}, \quad A := \Gamma_0(t \leftarrow s) = \begin{pmatrix} 0.58 & 0.42 \\ 0.42 & 0.58 \end{pmatrix},$$

so $CE = \left(\begin{smallmatrix} 0.62 & 0.38 \\ 0.38 & 0.62 \end{smallmatrix}\right) \neq A$ and, with $Z := \left(\begin{smallmatrix} 1 & -1 \\ -1 & 1 \end{smallmatrix}\right)$ (operator norm 2), $X = A - CE = -0.04\, Z$ and $D(t, s, r; \Gamma_0) = 0.08$. The reference is the symmetric two-state Glauber semigroup with rate λ, $\Gamma_{\mathrm{ref}}(\tau) = \frac{1}{2}\left(\begin{smallmatrix} 1+e^{-2\lambda\tau} & 1-e^{-2\lambda\tau} \\ 1-e^{-2\lambda\tau} & 1+e^{-2\lambda\tau} \end{smallmatrix}\right)$, divisible with stationary distribution $(\frac{1}{2}, \frac{1}{2})$; on the unit subintervals $D = F = \Gamma_{\mathrm{ref}}(1)$.

Substrate-dominant regime ($\lambda = 0.5$). Here $\Gamma_{\mathrm{ref}}(1) \approx \left(\begin{smallmatrix} 0.684 & 0.316 \\ 0.316 & 0.684 \end{smallmatrix}\right)$, $C - D = 0.01606\, Z$, $F - E = -0.11606\, Z$, and since $Z^2 = 2Z$, $K = (C-D)(F-E) = -0.00373\, Z$, so $M(t, s, r) \approx 0.00746$ and $K = cX$ with $c \approx 0.0932$. The identity gives $\Delta(t, s, r; \Gamma_g) = (1 - g)(-0.04 + 0.00373\, g)\, Z$; at $g = 0.5$ this predicts the entry -0.0191, and direct evaluation of $\Gamma_{0.5}(t \leftarrow s) - \Gamma_{0.5}(t \leftarrow r)\Gamma_{0.5}(r \leftarrow s)$ gives -0.019 to display precision. The exact response $D(t, s, r; \Gamma_g) = 2(1 - g)(0.04 - 0.00373\, g)$ is strictly decreasing on $[0, 1]$, so the two thresholds coincide; at $\varepsilon = 0.04$ the common value is $g_{\mathrm{first}} = g_{\mathrm{term}} \approx 0.477$. The quadratic bound of Theorem 4.3 gives $g_1 \approx 0.523$ and the linear bound $g_0 \approx 0.543$.

Interaction-dominant regime ($\lambda = 2$). Now $\Gamma_{\mathrm{ref}}(1) \approx \left(\begin{smallmatrix} 0.509 & 0.491 \\ 0.491 & 0.509 \end{smallmatrix}\right)$, $K \approx -0.111\, Z$, $M(t,s,r) \approx 0.222 > D(t,s,r;\Gamma_0) = 0.08$, and $K = cX$ with $c \approx 2.78$. The bound in (7) rises near $g = 0$ (Remark 3.3), while the exact response, $D(t,s,r;\Gamma_g) = 2(1-g)\,|0.04 - 0.111\, g|$, is non-monotone in the interior: it falls to a first crossing of $\varepsilon = 0.05$ at $g_{\mathrm{first}} \approx 0.108$, vanishes at $g^* = 1/c = 0.04/0.111 \approx 0.360$—the interior divisible coupling of Proposition 3.2(iii), where $X = g^*K$—then rebounds to a peak of ≈ 0.023 near $g \approx 0.68$ and descends to zero at $g = 1$. The peak stays below 0.05, so no re-entry occurs and $g_{\mathrm{term}} = g_{\mathrm{first}} \approx 0.108$; at any tolerance $\varepsilon < 0.023$ the rebound would re-cross, the thresholds would separate, and only g_{term} would certify permanent suppression. The bounds are valid but loose here—$g_1 \approx 0.807$ and $g_0 \approx 0.834$—because the interaction term is aligned with the substrate deficit and the sign-blind summaries cannot see it (Remark 4.5). The cone-geometric characterization of divisibility for two-state stochastic-quantum dynamics in [39] offers an alternative description of this regime; its relation to the operator-norm thresholds is an open question.

Attainment of the quadratic envelope ($\lambda = 0.35$). With the Glauber reference at rate λ and $q := \frac{1}{2}(1 - e^{-2\lambda})$, the unit-interval reference kernel has off-diagonal entry q, so $C - D = (q - 0.3)\, Z$, $F - E = (0.2 - q)\, Z$, and $K = 2(q - 0.3)(0.2 - q)\, Z$. Since $X = -0.04\, Z$, K points against X exactly when $q \in (0.2, 0.3)$, that is, $\lambda \in (0.2554, 0.4581)$; there $K = -cX$ with $c = M/D$, the configuration of Remark 4.5, and $D(t,s,r;\Gamma_g) = (1-g)(D + gM) = E(g)$ for every g. At $\lambda = 0.35$, $q = 0.2517$ and $M = 0.00999$; at $\varepsilon = 0.04$ the thresholds coincide with the quadratic bound, $g_{\mathrm{first}} = g_{\mathrm{term}} = g_1 = 0.5311$, against the linear bound $g_0 = 0.5555$.

Proposition 4.10 (Interior divisibility and affine extrapolation)**.** *Let Γ_{ref} be a divisible reference family and $g^* \in (0,1)$. The mixture Γ_{g^*} of Definition 2.6 is divisible if and only if*

$$\Gamma_0 \;=\; \frac{\Gamma_a - g^*\,\Gamma_{\mathrm{ref}}}{1 - g^*} \tag{14}$$

for a divisible two-time family Γ_a with $\Gamma_a(t \leftarrow s) \geq g^*\,\Gamma_{\mathrm{ref}}(t \leftarrow s)$ entrywise for all (t,s); then $\Gamma_{g^*} = \Gamma_a$. In that case $X = g^*K$ at every triple, $S_{\delta t}(g) = |g^* - g|\, M_*(\delta t)$, and

$$D_{\mathrm{CK}}(\delta t;\Gamma_g) \;=\; (1-g)\,|g^* - g|\, M_*(\delta t), \qquad g \in [0,1],\ \delta t \in (0,T]. \tag{15}$$

Proof. If Γ_{g^*} is divisible, put $\Gamma_a := \Gamma_{g^*}$; then $\Gamma_a - g^*\Gamma_{\mathrm{ref}} = (1-g^*)\Gamma_0$ is entrywise nonnegative and (14) holds. Conversely, if Γ_a is divisible with $\Gamma_a \geq g^*\Gamma_{\mathrm{ref}}$ entrywise, the right-hand side of (14) is entrywise nonnegative, has unit column sums, equals I at $t = s$, and is norm-continuous, so it is a two-time family in the sense of Definition 2.1, and $(1-g^*)\Gamma_0 + g^*\Gamma_{\mathrm{ref}} = \Gamma_a$ is divisible. When Γ_{g^*} is divisible, (6) gives $(1-g^*)[X - g^*K] = 0$ at every triple, so $X = g^*K$; then $\|X - gK\| = |g^* - g|\ \|K\|$, the supremum over $\mathcal{T}(\delta t)$ gives $S_{\delta t}(g) = |g^* - g|\, M_*(\delta t)$, and (15) is Proposition 3.2(i). □

Equation (14) is the global form of the pointwise proportionality in Proposition 3.2(iii): the mixture is divisible at an interior weight exactly when the substrate lies on the line through the reference and a divisible family, beyond that family.

A scale-level instance. Write a symmetric two-state kernel as $\frac{1}{2}\left(\begin{smallmatrix} 1+m & 1-m \\ 1-m & 1+m \end{smallmatrix}\right)$; composition multiplies m, so a time-homogeneous family of this form is divisible if and only if m is multiplicative in the duration. Take Γ_{ref} the Glauber semigroup with $\lambda = 2$, $m_{\mathrm{ref}}(\tau) = e^{-4\tau}$; Γ_a the Glauber semigroup with $\lambda = 1$, $m_a(\tau) = e^{-2\tau}$; and $g^* = 0.3603$. The substrate (14) has

$$m_0(\tau) \;=\; \frac{e^{-2\tau} - g^* e^{-4\tau}}{1 - g^*}, \qquad \tau = t - s \geq 0.$$

The numerator decreases from $1-g^*$ toward 0, its derivative $2e^{-2\tau}(2g^*e^{-2\tau}-1)$ being negative because $2g^*<1$, so $0<m_0\le 1$: Γ_0 is column-stochastic with $\Gamma_0(0)=I$, the entrywise condition of Proposition 4.10 holds, and m_0 is not multiplicative, so Γ_0 is indivisible at every scale. For a triple with early duration τ_1 and late duration τ_2, $\|X\| = |m_0(\tau_1+\tau_2) - m_0(\tau_1)m_0(\tau_2)|$ and $\|K\| = a(\tau_1)a(\tau_2)$ with $a(\tau) := m_0(\tau) - m_{\mathrm{ref}}(\tau) = (e^{-2\tau}-e^{-4\tau})/(1-g^*)$, which is maximal at $\tau = \frac{1}{2}\ln 2$ with value $1/[4(1-g^*)]$. Hence, for every $\delta t \ge \ln 2$,

$$M_*(\delta t) \;=\; \frac{1}{16(1-g^*)^2} \;=\; 0.152731, \qquad D_{\mathrm{CK}}(\delta t;\Gamma_0) \;=\; g^*M_*(\delta t) \;=\; 0.055029,$$

and (15) reads $D_{\mathrm{CK}}(\delta t;\Gamma_g) = 0.152731\,(1-g)\,|0.3603-g|$. At the scale level this family has the interior zero at $g^*=0.3603$, a rebound to $\frac{1}{4}(1-g^*)^2M_* = 1/64 = 0.015625$ at $g=(1+g^*)/2 = 0.6802$, and, for every tolerance below the rebound peak, separated thresholds: at $\varepsilon = 0.01$, $g_{\mathrm{first}} = 0.2705$ and $g_{\mathrm{term}} = 0.8721$, the deficit lying above ε again on $(0.4882, 0.8721)$, against the bounds $g_1 = 0.9500$ and $g_0 = 0.9519$.

5 The integrated deficit under perturbation

The companion paper consolidates the two-time family into the retarded transition operator $\mathcal{R}$ on $\mathcal{H}_T = L^2([0,T]\times\Omega)$ [1, Definition 7.1], a causal (filtration-adapted) operator [1, Proposition 7.3], and characterizes divisibility by the vanishing of the integrated kernel deficit

$$J(\Gamma) \;=\; \int_0^T\!\!\int_0^T\!\!\int_0^T \mathbf{1}_{\{s<r<t\}}\, D(t,s,r;\Gamma)^2\, ds\, dr\, dt \tag{16}$$

[1, Theorem 7.4], whose hypotheses (C1)–(C2) every family here satisfies. The retarded operator of the mixed family is the convex combination $\mathcal{R}_g = (1-g)\mathcal{R}_0 + g\,\mathcal{R}_{\mathrm{ref}}$ and is causal because each summand is; this is the only sense in which the causal organization of [1] enters, and the result below uses only the pointwise deficit.

Theorem 5.1 (Quadratic suppression of the integrated deficit)**.** *Under the hypotheses of Theorem 3.1, for each $g\in[0,1]$,*

$$\sqrt{J(\Gamma_g)} \;\le\; (1-g)\sqrt{J(\Gamma_0)} + g(1-g)\sqrt{J_M}, \qquad J_M := \int_0^T\!\!\int_0^T\!\!\int_0^T \mathbf{1}_{\{s<r<t\}}\, M(t,s,r)^2\, ds\, dr\, dt \;\le\; \frac{16N^2T^3}{6}, \tag{17}$$

so $J(\Gamma_g) \le (1-g)^2\big[\sqrt{J(\Gamma_0)} + g\sqrt{J_M}\big]^2 = O\big((1-g)^2\big)$ as $g\to 1$.

Proof. The pointwise bound (7) says $D(\cdot;\Gamma_g) \le (1-g)D(\cdot;\Gamma_0) + g(1-g)M(\cdot)$ on the simplex; taking L^2 norms over the simplex, monotonicity of the norm for nonnegative functions and Minkowski's inequality give (17). Finiteness: $M \le \|C-D\|\,\|F-E\| \le 4N$ by the $2\sqrt{N}$ bound on differences, and the simplex has volume $T^3/6$. The rate is an upper rate, because J is the square of an L^2 norm and the pointwise bound is $O(1-g)$; cancellation in (6) can make $J(\Gamma_g)$ decay faster, so no exact rate is asserted. (Squaring the pointwise bound with $(a+b)^2 \le 2a^2+2b^2$ gives the weaker $J(\Gamma_g) \le 2(1-g)^2J(\Gamma_0) + 2g^2(1-g)^2J_M$.) □

6 State-dependent coupling and monotone reparametrization

6.1 State-dependent coupling: a rigorous deficit bound

Let $g\colon [0,T]^2_{\le}\times\Omega \to [0,1]$ be any function, with $g_{\min} := \inf g$, $g_{\max} := \sup g$, and $\delta_g := g_{\max} - g_{\min}$. The state-dependent perturbation couples column j at weight $g(t,s,j)$:

$$\Gamma_{g(\cdot)}(t\leftarrow s)_{ij} \;=\; \big(1-g(t,s,j)\big)\,\Gamma_{0,ij}(t\leftarrow s) \;+\; g(t,s,j)\,\Gamma_{\mathrm{ref};ij}(t\leftarrow s). \tag{18}$$

Each column is a convex combination of the corresponding columns of Γ_0 and Γ_{ref}, so each $\Gamma_{g(\cdot)}(t \leftarrow s)$ is column-stochastic with $\Gamma_{g(\cdot)}(s \leftarrow s) = I$. This is a pointwise construction: it is a two-time family in the sense of Definition 2.1 only when g is continuous, in which case the scale-level supremum is attained as in Lemma 2.3; the bound below needs no regularity of g.

Proposition 6.1 (State-dependent coupling)**.** *With* $G(t,s) := \mathrm{diag}\big(g(t,s,1),\dots,g(t,s,N)\big)$,

$$\Gamma_{g(\cdot)}(t \leftarrow s) \;=\; \Gamma_0(t \leftarrow s)\,\big(I - G(t,s)\big) \;+\; \Gamma_{\mathrm{ref}}(t \leftarrow s)\,G(t,s), \tag{19}$$

and for every triple $0 \le s < r < t \le T$,

$$D(t,s,r;\Gamma_{g(\cdot)}) \;\le\; (1-g_{\min})\,D(t,s,r;\Gamma_0) \;+\; g_{\max}(1-g_{\min})\,M(t,s,r) \;+\; R(\delta_g), \tag{20}$$

with

$$R(\delta_g) \;=\; \frac{\delta_g}{2}\,\big(1{+}2\sqrt{N}\big)\,W_* + \Big(\frac{\delta_g\,W_*}{2}\Big)^2, \qquad W_* \;:=\; \sup_{(t,s)\in[0,T]^2_{\le}} \|\Gamma_{\mathrm{ref}}(t \leftarrow s) - \Gamma_0(t \leftarrow s)\|_{\mathrm{op}} \;\le\; 2\sqrt{N}. \tag{21}$$

Thus $R(\delta_g) = O(\delta_g)$ *uniformly over triples, the bound passes to any supremum over triples, and for constant weight* ($\delta_g = 0$) *it reduces to the bound* (7) *of Theorem 3.1.*

Proof. The factorization (19) is entrywise identical to (18), since right multiplication by the diagonal $G(t,s)$ rescales column j by $g(t,s,j)$. Fix $\bar g := (g_{\min} + g_{\max})/2$ and let $\Gamma_{\bar g}$ be the constant-weight perturbation. From (19), $P(t,s) := \Gamma_{g(\cdot)}(t \leftarrow s) - \Gamma_{\bar g}(t \leftarrow s) = \big(\Gamma_{\mathrm{ref}}(t \leftarrow s) - \Gamma_0(t \leftarrow s)\big)(G(t,s) - \bar g I)$, and since the operator norm of a diagonal matrix is its largest absolute entry, $\|P(t,s)\| \le W_* \max_j |g(t,s,j) - \bar g| \le W_*\delta_g/2 =: \rho$. With P_{ts}, P_{tr}, P_{rs} the perturbations on the three subintervals,

$$\Delta(t,s,r;\Gamma_{g(\cdot)}) \;=\; \Delta(t,s,r;\Gamma_{\bar g}) + P_{ts} - P_{tr}\,\Gamma_{\bar g}(r \leftarrow s) - \Gamma_{\bar g}(t \leftarrow r)\,P_{rs} - P_{tr}P_{rs},$$

and since constant-weight matrices are column-stochastic with operator norm at most $\sqrt{N}$, $D(t,s,r;\Gamma_{g(\cdot)}) \le D(t,s,r;\Gamma_{\bar g}) + \rho(1+2\sqrt{N}) + \rho^2$. Theorem 3.1 at weight $\bar g$ and the monotone replacements $1-\bar g \le 1-g_{\min}$, $\bar g(1-\bar g) \le g_{\max}(1-g_{\min})$ give (20). □

Remark 6.2 (Branch mixtures produce state-dependent weights). The most natural stochastic construction produces the state-dependent family. Suppose two processes Y and Z on Ω have two-time kernels Γ_{ref} and Γ_0 and one-time laws π_s^Y, π_s^Z, and let B be an independent Bernoulli(g) variable selecting Y on $\{B=1\}$ and Z on $\{B=0\}$. For states j of positive probability under the mixture, conditioning on $X_s = j$ updates the branch weight by Bayes' rule, so the two-time kernel of the mixture is (18) with the weight

$$g(t,s,j) \;=\; \frac{g\,\pi_s^Y(j)}{g\,\pi_s^Y(j) + (1-g)\,\pi_s^Z(j)},$$

which depends on s and j and is constant in t; it reduces to the fixed weight g of Definition 2.6 exactly when $\pi_s^Y = \pi_s^Z$ for all s. At a state with $\pi_s^Y(j) = \pi_s^Z(j) = 0$ the conditional kernel is undetermined; completing that column with the constant weight g keeps the family in the class (18) and changes neither δ_g nor the estimate of Proposition 6.1, because at each s both one-time laws sum to one over the states of positive probability, so some state has $\pi_s^Y(j) \ge \pi_s^Z(j)$ and some has $\pi_s^Y(j) \le \pi_s^Z(j)$, and g lies between the smallest and the largest Bayes weight. For this reason the fixed-weight model is presented as a formal perturbation path rather than as the kernel of a specified protocol (Remark 2.7), and Proposition 6.1, with δ_g controlled by the disparity of the one-time laws, is the estimate that applies to the branch mixture. Whether Γ_0 itself is process-induced is a separate question that the theorems do not require.

6.2 Monotone reparametrization of the coupling

Replacing g by $f(g)$ for a continuous nondecreasing $f\colon [0,1] \to [0,1]$ with $f(0) = 0$, $f(1) = 1$ is a reparametrization of the same affine family, not a new coupling; the kernels remain affine in the reparametrized weight. Its only content is how thresholds transform. Since f need not be injective, they transform through the generalized inverse $f^{\leftarrow}(y) := \inf\{g \in [0,1] : f(g) \geq y\}$ rather than f^{-1}: with g_1 the sufficient coupling of Theorem 4.3, for every $\gamma \in (g_1, 1]$ and every $g \geq f^{\leftarrow}(\gamma)$, monotonicity and continuity of f give $f(g) \geq \gamma > g_1$, hence $D_{\mathrm{CK}}(\delta t; \Gamma_{f(g)}) < \varepsilon$ by Theorem 4.3(ii), so the terminal threshold in the g-parametrization satisfies $g^{(f)}_{\mathrm{term}}(\delta t; \varepsilon) \leq f^{\leftarrow}(\gamma)$ for every $\gamma \in (g_1, 1]$, collapsing to $g^{(f)}_{\mathrm{term}} \leq f^{-1}(g_1)$ when f is strictly increasing. The two-threshold structure is invariant under such reparametrizations.

7 Discussion

7.1 Divisibility-based non-Markovianity functionals

Remark 7.1. The divisibility-based functionals of Wolf–Eisert–Cubitt–Cirac [10], Rivas–Huelga–Plenio [12, 13], Chruściński–Maniscalco [14], and Buscemi–Datta [15] are defined on quantum dynamical maps; the joint quantum/classical formulation of Vacchini et al. [16] and the P-divisibility transfer of Benatti–Chruściński–Nichele [17] bridge to the column-stochastic setting. The threshold behavior established here *may* induce a corresponding, tolerance-dependent behavior of such a functional, but a transfer requires a continuity assumption or an explicit comparison inequality between the functional and D_{CK}, which we do not supply; no consequence for any functional is asserted. By the Buscemi–Datta equivalence between divisibility and monotonic decrease of information, exact monotonicity of every data-processing-compatible measure holds exactly where the mixed family is divisible: guaranteed at $g = 1$, and by Proposition 3.2(iii) at most at one interior coupling, as the zero at $g^* = 0.3603$ in Section 4.3 illustrates, pointwise and at the scale level. Above g_{term} the deficit is below ε but need not vanish, so the terminal threshold marks tolerance-level suppression of the deficit, not restoration of information monotonicity.

7.2 Process tensors

The process-tensor formalism [26, 5, 27, 28] treats memory in CPTP dynamics through a multi-time dilation over discrete sequences of control operations; its non-Markovianity is a Markov-order property of a multilinear object, not an operator-norm CK deficit of a continuous two-time family. The present results have no process-tensor analogue and no process-tensor specialization is claimed. The settings are compatible in one direction only: a column-stochastic family is a diagonal-CPTP specialization in the sense of the diagonal-retention lift [1, Definition 5.1].

7.3 Minimal Markovian admixture

The robustness of non-Markovianity of Anand–Brun [23] is the minimum Markovian noise that must be mixed with an evolution to render it Markovian; De Santis–Giovannetti [24] pose the constant-weight incoherent-mixing version and minimize the weight subject to Markovianity at all times; Bhattacharya–Bhattacharya–Majumdar [25] give the resource-theoretic framing with Markovian semigroups as free operations. The terminal suppression threshold belongs to this family of ideas: it is a fixed-reference, tolerance-relaxed variant of the De Santis–Giovannetti measure. The differences are those listed in the introduction: the reference here is fixed and the threshold is in the weight against it, not a minimization over admixtures; the object is

the classical two-time CK deficit in operator norm rather than CP-divisibility; suppression is to tolerance ε up to a temporal scale rather than exact Markovianization at all times; and the exact identity, with its asymmetric interaction term and the first-crossing/terminal distinction, has no counterpart when both mixed dynamics are general. A formal connection between the two families of results through the diagonal-CPTP specialization is plausible and is not attempted here.

7.4 Product-form inequalities

The product form $g_{\mathrm{first}}(\delta t;\varepsilon)\,\delta t \geq \eta(\varepsilon)\,\delta t_-$ of Proposition 4.8 resembles the product-form inequalities of speed-limit theory—Mandelstam–Tamm [29], Margolus–Levitin [30], their classical-stochastic analogues [31, 32, 33], the Fisher-information unification of Nishiyama–Hasegawa [36], and the driven-system and review literature [34, 35]. The resemblance is formal only: the inequality arises by multiplying a uniform bound on a mixing weight by the endpoint of a chosen band, its constant is not universal, g is not a rate or a statistical distance, and no $\delta t \to 0$ asymptotics or path-length geometry is involved. We do not pursue a rate-form or information-geometric formulation, and we withdraw the question, posed in an earlier version of this paper, of whether one exists.

7.5 Scope of the detectability notion

Detectability as defined in Section 4 is deterministic and population-level: D_{CK} is a property of the family, and ε is a prescribed tolerance, not a significance level. From finite transition data, $D_{\mathrm{CK}}(\delta t;\Gamma_g)$ is estimated with sampling error, and "below ε" becomes a statistical statement requiring an estimator of the deficit, its finite-sample distribution under the null hypothesis of a divisible family, and a calibration of ε against that distribution; none of this is supplied here, and "indistinguishable from a divisible family" throughout means indistinguishable according to the CK-deficit statistic at the chosen temporal scale and tolerance. The scope is restricted in two further respects. The mixture modifies the transition family by definition; whether a physical intervention modifies the underlying dynamics, the observation protocol, or both is not derived, and no claim about observers or apparatus is made. The deficit subspaces of [1, Theorem 4.5] computed for the substrate are independent of g, whereas those of the mixed family are not: by (6), $\Delta(t,s,r;\Gamma_g)$ is proportional to $X - gK$, whose range and corange move with g. The two-threshold theory establishes a quantitative version of the structural point of [1, 2]: whether the two-time CK deficit is present at a given scale and tolerance is a property of the (substrate, reference, weight, scale, tolerance) quintuple, and the exact identity makes the dependence on the weight computable.

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